

CS671 – Deep Learning and Its Applications

Technical Documentation Report

Model: Multi-Model Comparative Analysis

Project Group: 5 Region: Brain/Pelvis Modality: Mixed

1. ARCHITECTURAL COMPARISON

We analyze the trade-offs between 8 implemented model variants, categorized by their supervision requirements (Paired vs. Unpaired) and generative paradigms (GAN vs. Diffusion).

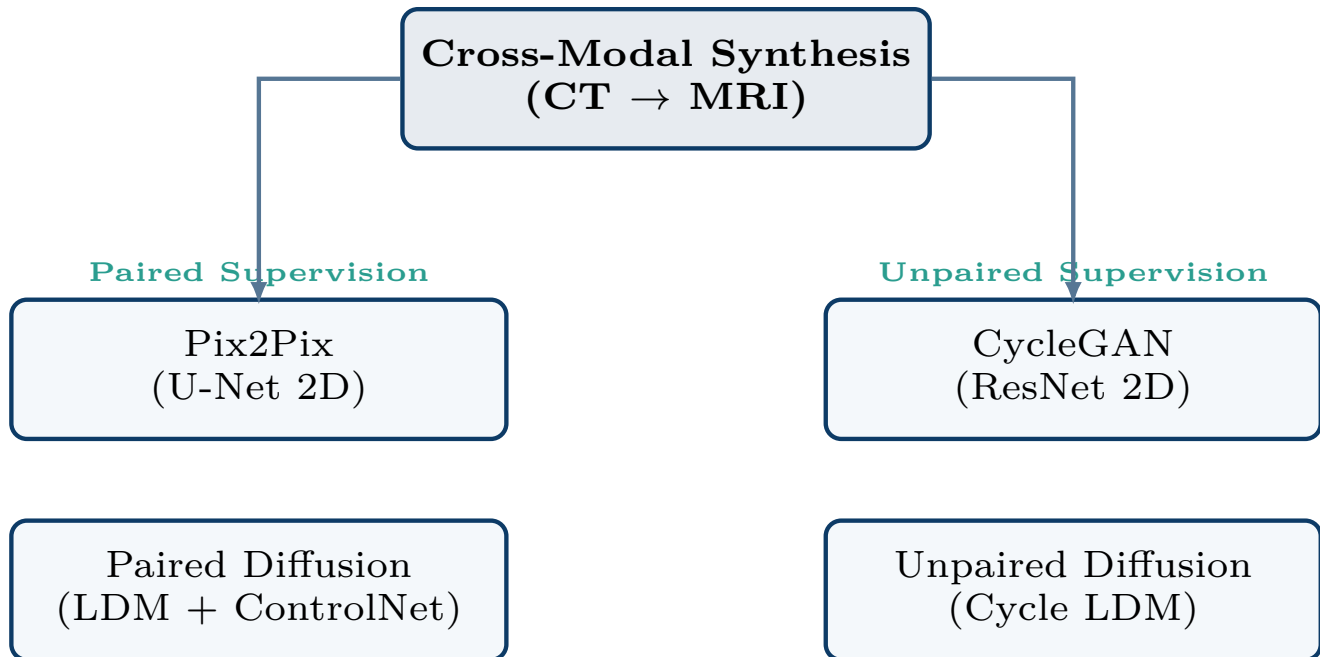


Figure 1: Systematic taxonomy of implemented CT-to-MRI translation pipelines.

Paired Paradigm (Direct Mapping): Pix2Pix and Paired Diffusion rely on rigidly aligned CT-MRI pairs. Pix2Pix uses a U-Net architecture that directly minimizes L_1 pixel distance, leading to sharp but occasionally over-smoothed results. Paired Diffusion, using SD-Turbo as a backbone, leverages pre-trained generative priors to synthesize high-frequency textures that are statistically consistent with MRI manifolds.

Unpaired Paradigm (Distribution Alignment): CycleGAN and Unpaired Diffusion operate on independent CT and MRI datasets. The "Cycle-Consistency" constraint serves as a surrogate for spatial alignment, forcing the model to learn bijective mappings. This is particularly effective for the Pelvis region, where clinical scans often have significant intra-patient anatomical variations due to bladder filling or bowel gas.

2. MODEL SPECIFICATIONS

Feature	Pix2Pix	CycleGAN	Paired Diffusion	Unpaired Diffusion
Loss Function	$L_{adv} + \lambda L_1$	$L_{adv} + L_{cyc} + L_{idt}$	ϵ -MSE + LPIPS	Latent-Cycle + L_{adv}
Generative Engine	U-Net-256	ResNet-9	UNet (Latent)	Cycle-UNet (Latent)
Normalization	Batch Norm	Instance Norm	Group Norm	Group Norm
Advantage	High Fidelity	No Alignment Needed	Multi-modal Priors	Best of Both Worlds
Disadvantage	Artifacts on Shift	Hallucinations	Slow Inference	Training Instability

Table 1: Comparative analysis of architectural and optimization specifications.

3. TECHNICAL DISCUSSION

The comparative study reveals that while **Pix2Pix** is the most efficient for Brain scans (where alignment is near-perfect), **Diffusion-based** models offer superior perceptual quality (SSIM and LPIPS) by operating in the latent space. The use of **2.5D representations** (conditioning on adjacent slices) in the Diffusion models significantly reduces axial discontinuity, a common problem in purely 2D GAN-based reconstruction. For Pelvis synthesis, the

Cycle-Consistent Diffusion variant emerges as the most robust architecture, balancing the texture realism of diffusion with the structural constraints of CycleGAN.